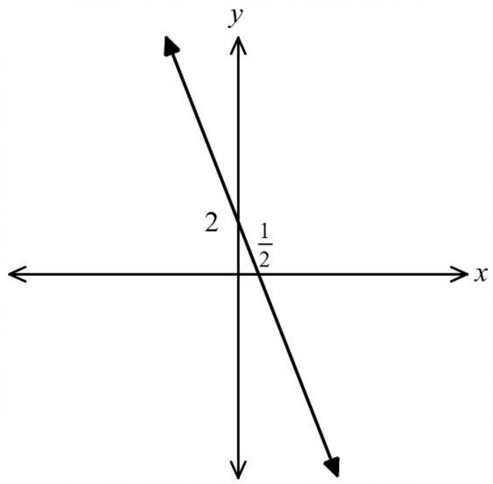
Western Mathematics Exams

2022
YEAR 11
EXAMINATION

Mathematics Standard

SOLUTIONS

Multiple Choice Worked Solutions

No	Working	Answer
1	Price after Profit margin $= \$120 \times 140\%$ $= \$120 \times 1.4 = \$168$ Price after GST added $= \$168 \times 1.1 = \$184.80$ Or Selling Price $= \$120 \times 1.4 \times 1.1 = \$184.80$	**D**
2	$y = -4x + 2$ has a gradient of -4 and a y-intercept of 2.  	**D**
3	0.003 567 92 when rounded to 3 significant figures is 0.00357 To write in S N we need a number between 1 and 10 times a power of 10. So we get 3.57×10^{-3} as we need to move the decimal point three places to the left to get our required number of 0.00357.	**D**
4	Area of rhombus = area of parallelogram $= b \times h$ $= 4 \times 5 = 20\text{ cm}^2$ Area of two semicircles with 5 cm diameter = area of circle with 2.5 cm radius $= \pi \times 2.5^2$ $= 19.63495408$ $= 19.6\text{ cm}^2$ (to 1 decimal place) Area of shape $= 20 + 19.6$ $= 39.6\text{ cm}^2$ (to 1 decimal place)	**B**
5	Commission on sales up to \$24 000 $= 24000 \times 0.03 = \$720$ Commission on sales over \$24 000 $= (36000 - 24000) \times 0.05 = \$600$ Pay for the week = retainer + commission $= 500 + 720 + 600$ $= \$1\,820.00$	**C**
6	The median is the middle of the dataset. Where there is an even number of scores, the median is the average of the two middle scores, in this case the 12th and 13th scores. 0 | 4 5 8 9 1 | 1 1 3 5 6 8 2 | 2 6 8 8 9 9 9 3 | 0 2 2 4 4 | 0 2 4 The average of 26 and 28 $= \frac{26+28}{2} = 27$	**B**

<table>
<tr><td>7</td><td>Water usage of 60 kL is a reduction of 12 kL
Reduction in charge $= 12 \times 2.10 = \$25.20$.
New total $= 322.18 - 25.20 = \$296.98$</td><td>C</td></tr>
<tr><td>8</td><td>Total with mild to Severe $= 0.06 + 0.02 + 0.07 = 0.15$
Expected frequency in 85 000 $= 85000 \times 0.15 = 12\,750$</td><td>A</td></tr>
<tr><td>9</td><td>$$\begin{aligned} \text{Childs dose} &= \frac{\text{age(in months)} \times \text{adult dosage}}{150} \\ 6 &= \frac{18 \times \text{adult dosage}}{150} \\ 6 \times 150 &= 18 \times \text{adult dosage} \\ \text{adult dosage} &= \frac{6 \times 150}{18} \\ &= 50 \text{ millilitres} \end{aligned}$$</td><td>B</td></tr>
<tr><td>10</td><td>The median of 18 scores is the average of the 9th and 10th scores (both 12)= 12
Lower quartile = middle of lower 9 scores, so the 5th score =10
Upper quartile = middle of upper 9 scores, so 14th score =13
<table>
<tr><td></td><td></td><td></td><td></td><td>O</td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td>O</td><td>O</td><td></td><td>O</td><td></td></tr>
<tr><td></td><td>O</td><td>O</td><td>O</td><td>O</td><td>O</td><td>O</td><td></td></tr>
<tr><td>O</td><td>O</td><td>O</td><td>O</td><td>O</td><td>O</td><td>O</td><td>O</td></tr>
<tr><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td></tr>
</table>
The interquartile range = 13 – 10 = 3</td><td>A</td></tr>
</table>

2022 YEAR 11 Final Examination
Mathematics Standard

Name ______________________________ Teacher ______________________

Section I – Multiple Choice Answer Sheet

Allow about 25 minutes for this section
Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample:	2 + 4 =	(A) 2	(B) 6	(C) 8	(D) 9
		A ○	B ●	C ○	D ○

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

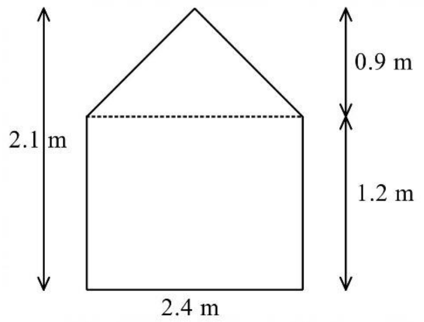
A ● B ~~●~~ C ○ D ○

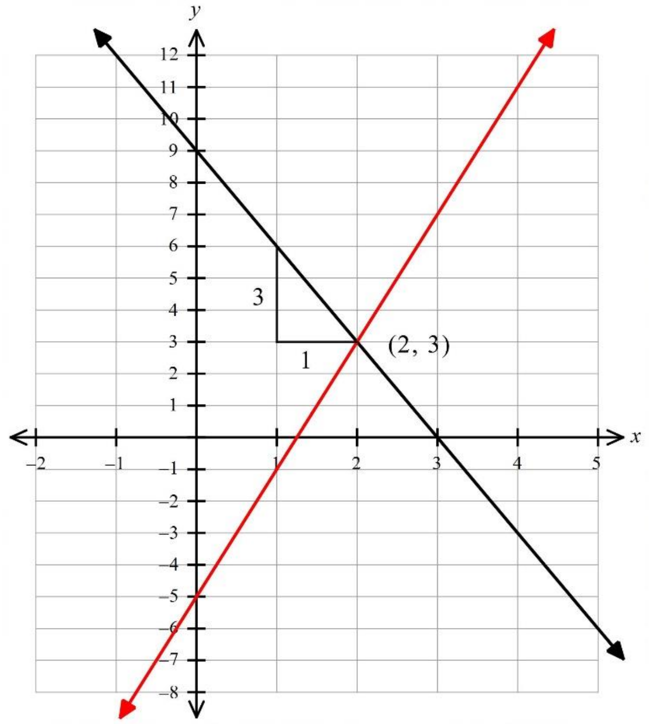
If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ~~●~~ B ~~●~~ correct C ○ D ○

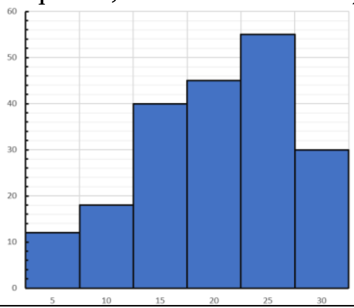
1.	A ○	B ○	C ○	D ●
2.	A ○	B ○	C ○	D ●
3.	A ○	B ○	C ○	D ●
4.	A ○	B ●	C ○	D ○
5.	A ○	B ○	C ●	D ○
6.	A ○	B ●	C ○	D ○
7.	A ○	B ○	C ●	D ○
8.	A ●	B ○	C ○	D ○
9.	A ○	B ●	C ○	D ○
10.	A ●	B ○	C ○	D ○

	Solution	Marks	Allocation of marks
Question 11			
	Suggested solution 1 $10-\frac{k}{3}=k+2$ $3\times 10-3\times\frac{k}{3}=3\times k+3\times 2$ $30-k=3k+6$ $30=4k+6$ $24=4k$ $k=\frac{24}{4}=6$ Suggested solution 2 $10-\frac{k}{3}=k+2$ $8-\frac{k}{3}=k$ $3\times 8-3\times\frac{k}{3}=3\times k$ $24-k=3k$ $24=4k$ $k=\frac{24}{4}=6$	2	2 marks for correct answer 1 mark for some correct and relevant working Other methods are possible, including guess and check
Question 12			
	$\text{Absolute error} = \frac{1}{2}\times \text{smallest measurement}$ $= \frac{1}{2}\times 0.2\text{ kg}$ $= 0.1\text{ kg}$ $\text{Limits} = 5.0 \pm 0.1\text{ kg}$ $= 4.9 \text{ and } 5.1\text{ kg}$	2	2 marks for correct limits 1 mark for correct absolute error or equal merit
Question 13			
(a)	Highest maximum = 30 and lowest minimum = 5 Difference = 30 – 5 =25	1	1 mark for correct answer
(b)	The maximum temp varies from day to day, but the overall trend for the month is downward, i.e. the temperatures are getting lower.	1	1 mark for correct answer
Question 14			
	$s=ut+\frac{at^2}{2}$ $s=195,\ t=5,\ a=10,$ find u $s=ut+\frac{at^2}{2}$ $195=5u+\frac{10\times 5^2}{2}$ $195=5u+125$ $5u=195-125$ $5u=70$ $u=\frac{70}{5}=14$	2	2 marks for correct answer 1 mark for correct substitution with error in solution or equal merit

Question 15			
	Total Hours $= 7+8+7+8+7+4 = 41$ hours Pay for first 36 hours $= 36 \times 46.40 = \$1670.40$ Pay for next 4 hours $= 4 \times 46.40 \times 1.5 = \$278.40$ Pay for remaining hour $= 1 \times 46.40 \times 2 = \$92.80$ Pay for the week $= 1670.40 + 278.40 + 92.80 = \$2\,041.60$	2	2 marks for correct answer 1 mark for correct calculation of total hours and breaking into brackets with error in solution or equal merit
Question 16			
	The ends are pentagons that can be divided into a rectangle and a triangle.  Area of one End $= 2.4 \times 1.2 + \frac{1}{2} \times 2.4 \times 0.9$ $= 3.96 \text{ m}^2$ Area of Fabric $= 2 \times$ ends $+ 2 \times$ sides $+ 2 \times$ slopeing roof $= 2 \times 3.96 + 2 \times 2.5 \times 1.2 + 2 \times 2.5 \times 1.5$ $= 7.92 + 6 + 7.5$ $= 21.42 \text{ m}^2$	2	2 marks for correct answer 1 mark for correct calculation of ends or of total area with a single error or equal merit
Question 17			
	$\text{P(not \$8 or \$16)} = P(\text{Lose or \$4})$ $= \frac{16}{25} + \frac{7}{25}$ $= \frac{23}{25} = 0.92$ OR $\text{P(not \$8 or \$16)} = 1 - P(\text{\$8 or \$16})$ $= 1 - \left(\frac{3}{50} + \frac{1}{50}\right)$ $= 1 - \frac{4}{50}$ $= \frac{23}{25}$	2	2 marks for correct answer 1 mark for working including relevant probabilities

<table>
<tr><td colspan="2">Question 18</td><td colspan="2"></td></tr>
<tr><td>(a)</td><td>

Using any two points on the line.
e.g. $(1,\ 6)$ and $(2,\ 3)$
$m = \dfrac{6-3}{1-2} = \dfrac{3}{-1} = -3$</td><td>1</td><td>1 mark for correct gradient</td></tr>
<tr><td>(b)</td><td>The line will have a y-intercept of –5 and a gradient of 4.
This gives the second (red) line on the graph above.</td><td>1</td><td>1 mark for correct line</td></tr>
<tr><td>(c)</td><td>The point of intersection is (2, 3) or other if error made in (b).</td><td>1</td><td>1 mark for correct point, from the line drawn in (b).</td></tr>
</table>

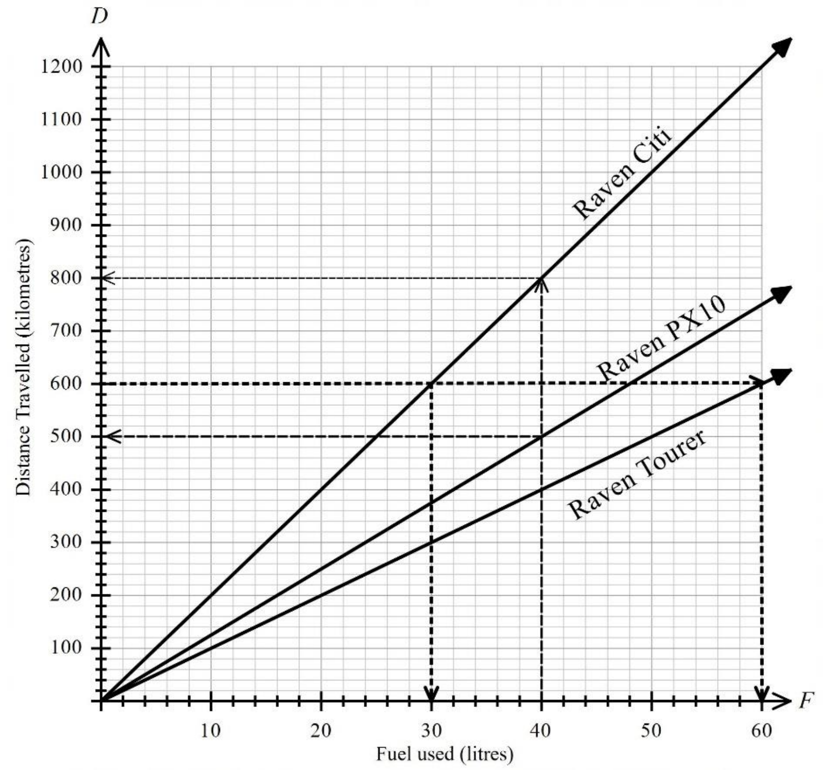
<table>
<tr><td colspan="2">Question 19</td><td colspan="2"></td></tr>
<tr><td>(a)</td><td>
<table>
<tr><th>Class</th><th>Class Centre</th><th>Frequency</th><th>Cumulative Frequency</th></tr>
<tr><td>2.5 – 7.5</td><td>5</td><td>12</td><td>12</td></tr>
<tr><td>7.5 – 12.5</td><td>10</td><td>18</td><td>30</td></tr>
<tr><td>12.5 – 17.5</td><td>15</td><td>40</td><td>70</td></tr>
<tr><td>17.5 – 22.5</td><td>20</td><td>45</td><td>115</td></tr>
<tr><td>22.5 – 27.5</td><td>25</td><td>55</td><td>170</td></tr>
<tr><td>27.5 – 32.5</td><td>30</td><td>30</td><td>200</td></tr>
</table>
</td><td>1</td><td>1 mark for correct line</td></tr>
<tr><td>(b)</td><td>Modal class = 22.5 – 27.5 (class with the highest frequency)</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td>(c)</td><td>$\text{Mean} = \dfrac{5 \times 12 + 10 \times 18 + 15 \times 40 + 20 \times 45 + 25 \times 55 + 30 \times 30}{200}$
$= \dfrac{4015}{200}$
$= 20.075$</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td>(d)</td><td>The data set is negatively skewed (a quick sketch, which is not required, will look like this)
</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td colspan="2">Question 20</td><td colspan="2"></td></tr>
<tr><td>(a)</td><td>Using similar triangles,
$\dfrac{x}{0.9} = \dfrac{1.6}{3.6}$
$x = \dfrac{0.9 \times 1.6}{3.6} = 0.4$

$\text{Area Trapezium} = \dfrac{2.0}{2}(0.9 + 0.4) = 1.3 \text{ m}^2$
$\text{Volume Solid} = 1.3 \times 0.8 = 1.04 \text{ m}^3$</td><td>2</td><td>2 marks for similar triangles proof and for volume

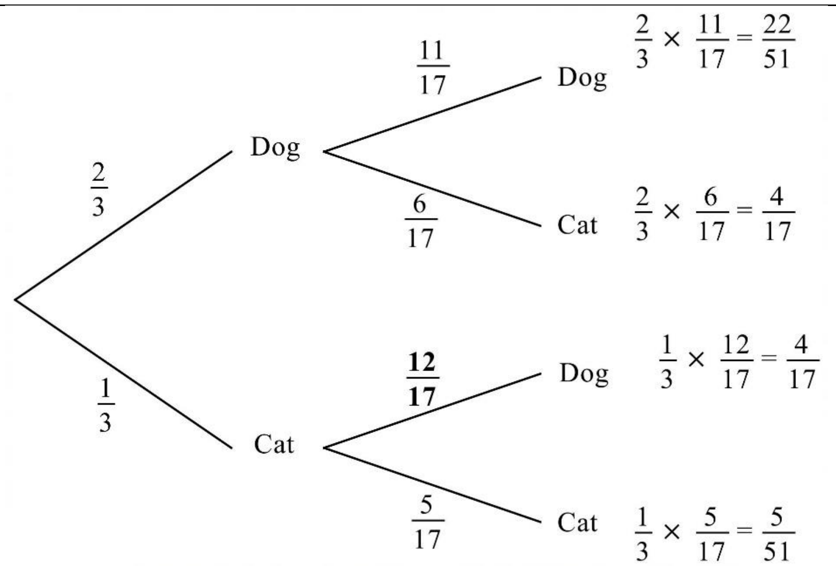
1 mark for either of proof or volume</td></tr>
<tr><td>(b)</td><td>$\text{Mass} = 1.04 \times 7850 = 8\,164 \text{ kg}$
$= 8.2 \text{ tonnes}$ (nearest 10th of a tonne)</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td colspan="2">Question 21</td><td colspan="2"></td></tr>
<tr><td>(a)</td><td>1500 w = 1.5 kW
In 24 hours the heater will use $1.5 \times 24 = 36$ kWh
Cost of heater $= 36 \times 0.234 = 8.424 = \$8.42$</td><td>2</td><td>2 marks for correct answer

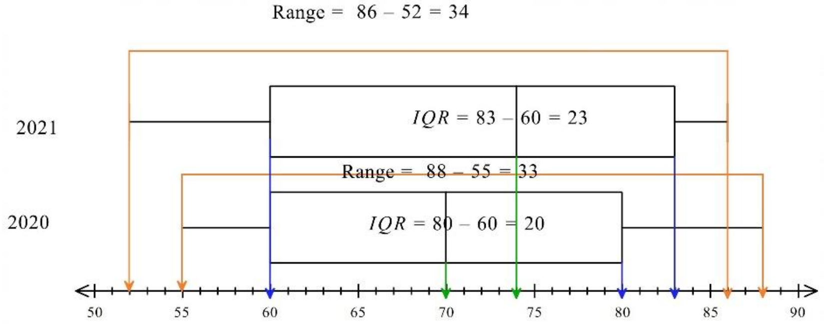
1 mark for correct number of kWh or equal merit</td></tr>
</table>

Question 22			
(a)	$\mathrm{Y} \times 12 = 385.20$ $\mathrm{Y} = \dfrac{385.20}{12} = \$32.10$ $\mathrm{Z} = 2\,704.80 + 1\,846.00 + 385.20 + 856.00 + 4\,472.00 + 1\,300.00 = \$11\,564.00$	2	2 marks for both Y and Z correct 1 mark for one of the values correct
(b)	$\text{Weekly amount} = \dfrac{11564.00}{52} = 222.38461538$ $= \$222.40$ (Nearest 10 c)	1	1 mark for the correct value or an answer found correctly from the answer in (a)
Question 23			



(a)	Reading from 40 litres, the PX10 travels 500 km and the Citi travels 800 km, so the Citi travels 300 km extra.	1	1 mark for correct answer
(b)	The line will begin at the origin. Find another point using the information of 10 L per 100 km, so could plot (10, 100) or for more accuracy another point such as (50, 500). Joint the points to draw the line and label the line as *Raven Tourer* as shown above.	2	2 marks for correct line 1 mark for some relevant working or an incompletely drawn line or equal merit
(c)	Following across from 600 km, we get 30 litres for the Citi, around 48 litres for the PX10 and 60 litres for the Tourer. The difference between the highest and lowest is 60 – 30 = 30 litres.	1	1 mark for correct answer

Question 24			
	Volume $= \frac{1}{2} \times \frac{4}{3}\,\pi\,r^3$ $= \frac{1}{2} \times \frac{4}{3} \times \pi \times 30^3$ $= 56548.66776462$ $= 56\,549\ \text{cm}^3$ Capacity in litres $= \frac{56549}{1000}$ $= 56.549$ $=$ 57 litres (nearest litre)	2	2 marks for correct answer 1 mark for correct volume or equal merit
Question 25			
(a)		1	1 mark for missing value of $\frac{12}{17}$
(b)	$P(DC) = \frac{2}{3} \times \frac{6}{17} = \frac{4}{17}$	1	1 mark for correct value
(c)	P(same species) $= P(DD) + P(\text{CC}) = \frac{22}{51} + \frac{5}{51}$ $= \frac{27}{51} = \frac{9}{17}$	1	1 mark for correct value simplified or not
Question 26			
(a)	Their combined annual salary = \$53 049 + \$44 980 = \$98 029 Their combined weekly salary = $\frac{\$98\,029}{52}$ = \$1 885.17 Their combined weekly saving = \$1885.17 − 1643 = \$242.17 Annual saving = 242.17 × 52 = \$12 592.84 If you calculate the annual saving first it is 12 593.00	2	2 marks for both correct answers 1 mark for one correct answer or valid working toward answers

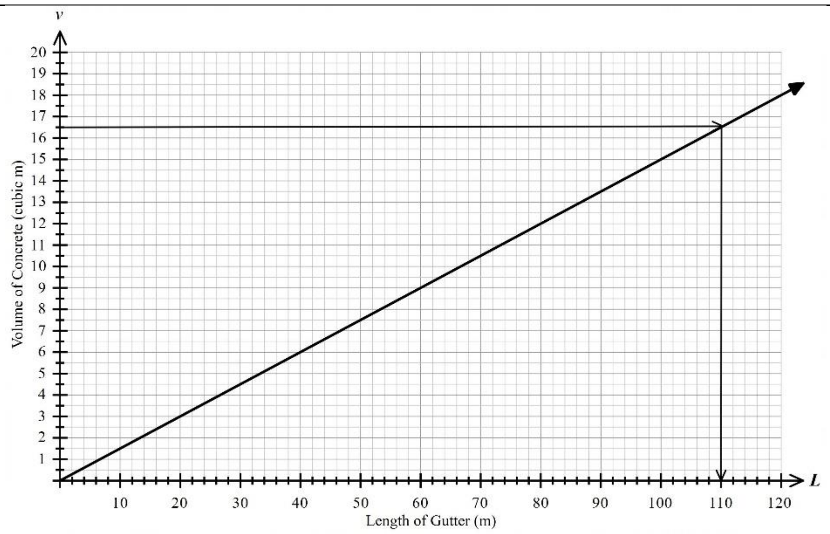
Question 27			
(a)	 HSC Marks in Standard Mathematics at Danville High The range (red) for 2021 is 34 compared to 33 for 2020, so 1 greater and the interquartile range (blue) is 23 for 2021 compared to 20 for 2020 which is 3 greater, so both the measures show that the spread was greater in 2021.	2	2 marks for correct comment which correctly quotes two measures of spread. 1 mark for a comment that correctly quotes one measure
(b)	The median measure (green) for 2021 is 74 compared to 70 for 2020, and the two have the same lower quartile, but the 2021 has an upper quartile which is greater by 3 marks, so both support the claim of an improvement from 2020 to 2021.	1	1 mark for a comment that quotes any measure that supports the claim.
Question 28			
	$B = P + \frac{PRN}{100}$ $B - P = \frac{PRN}{100}$ $100(B - P) = PRN$ $R = \frac{100(B - P)}{PN}$ or $R = \frac{100B - 100P}{PN}$	3	3 marks for correct formula in either form 2 marks for minor error in algebra or slightly incomplete solution 1 mark for some relevant working
Question 29			
	$V_0 = \$15\,600$ $S = V_0 - Dn$ $D = 15600 \times 0.0125 = 195$ (ie \$195 per month) $S = 5000$ $5000 = 15600 - 195n$ $5000 - 15600 = -195n$ $-10600 = -195n$ $n = \frac{-10600}{-195}$ $= 54.35897436$ It could be replaced after 55 months.	2	2 marks for correct answer 1 mark for correct value of D and attempt to find n or equal merit

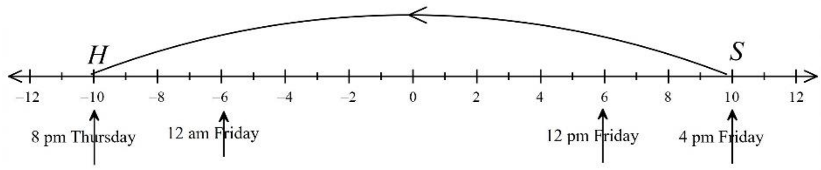
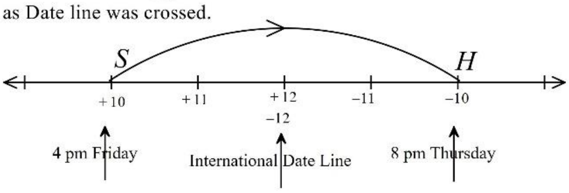
<table>
<tr><td colspan="2">Question 30</td><td colspan="2"></td></tr>
<tr><td>(a)</td><td>Total income $= 98020 + 560$
$= \$98\,580.00$
Total deductions $= \$2540 + 3200 + 895 + 3453$
$= \$10\,088.00$
Taxable income $= \$98580.00 - 10088.00$
$= \$88\,492.00$</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td>(b)</td><td>Income of $88 492.00 lies in the third bracket.
Income Tax $= \$5\,092 + 0.325 \times (88\,492 - 45\,000)$
$= \$19\,226.9$
Medicare $Levy = 0.02 \times 88492$
$= \$1769.84$
Total Tax $= \$19226.9 + 1769.84$
$= \$20\,996.74$</td><td>2</td><td>2 marks for correct answer

1 mark for correct income tax or correct Medicare Levy or equal merit</td></tr>
<tr><td>(c)</td><td>Total $PAYG$ tax paid $= \$825.60 \times 26 = \$21\,465.60$
Refund $= 21465.6 - 20996.74$
$= \$468.86$</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td colspan="2">Question 31</td><td colspan="2"></td></tr>
<tr><td>(a)</td><td>
<table>
<tr><td></td><td>Red</td><td>Green</td><td>Yellow</td><td>Purple</td><td>Ebony</td><td>Bald</td></tr>
<tr><td>Male Dwarf</td><td></td><td>GMD</td><td></td><td></td><td></td><td>BMD</td></tr>
<tr><td>Female Dwarf</td><td></td><td>GFD</td><td></td><td></td><td></td><td>BFD</td></tr>
<tr><td>Male Elf</td><td></td><td>GME</td><td></td><td></td><td></td><td></td></tr>
<tr><td>Female Elf</td><td></td><td>GFE</td><td></td><td>PFE</td><td></td><td></td></tr>
<tr><td>Male Hobbit</td><td>RMH</td><td>GMH</td><td>YMH</td><td>PMH</td><td>EMH</td><td>BMH</td></tr>
<tr><td>Female Hobbit</td><td>RFH</td><td>GFH</td><td>YFH</td><td>PFH</td><td>EFH</td><td>BFH</td></tr>
</table>

P(PFE) $= \frac{1}{36}$</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td>(b)</td><td>P(G or H) $= \frac{16}{36} = \frac{4}{9}$</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td>(c)</td><td>P(BD) $= \frac{2}{36} = \frac{1}{18}$
P(Not BD) $= 1 - \frac{1}{18} = \frac{17}{18}$</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="2">Question 32</td><td colspan="2"></td></tr>
<tr><td></td><td>Working backwards
Needs to be at appointment at 9:00
Needs to arrive at Karvalo Rd at 8:50 at latest
Last tram arrives at Karvalo Rd at 8:38
This Tram leaves Amstrong St at 7:56
Needs to leave home 15 min before 7:56, which is 7:41.</td><td>2</td><td>2 marks for correct answer

1 mark for incorrect answer with some relevant working/reasoning shown</td></tr>
</table>

Question 33			
(a)	Distance between offsets $= 72 \div 4 = 18$ m Use the formula $A \approx \dfrac{h}{2}\left(d_f + d_l\right)$ four times Area $= \dfrac{18}{2}(45+39) + \dfrac{18}{2}(39+35) + \dfrac{18}{2}(35+29) + \dfrac{18}{2}(29+39)$ $= 756 + 666 + 576 + 612$ $= 2\,610 \text{ m}^2$	2	2 marks for correct answer 1 mark for correct use of formula with an error
Question 34			
(a)	$v = kL$ $6 = k \times 40$ $k = \dfrac{6}{40} = 0.15$	1	1 mark for correct answer
(b)		2	2 marks for correct graph 1 mark for straight line graph with zero intercept but incorrect gradient or equal merit
(c)	Read off graph as shown above or use formula $v = 0.15L$ $16.5 = 0.15L$ $l = \dfrac{16.5}{0.15} = 110$ metres	1	1 mark for correct answer

Question 35			
	To get from +10 to –10, go backward 20 hours.  *OR* to get from +10 to –10, go forward 4 hours and back one day, as Date line was crossed.  She leaves at 8 pm on the previous Thursday Honolulu time. She calls her mum after $6+3+3 = 12$ hours later which is 8 am on Friday. Her mum received the call at 4 pm + 12 hours = 4 am on Saturday.	3	3 marks for correct time for Sydney and Honolulu. 2 marks for correct Honolulu time for call or equal merit. 1 mark for correct Sydney time or equal merit

Question 36

New Annual Income $= \$58760 \times 1.025 = 60229$

New Weekly Income $= = 60229 \div 52 = \$1158.25$

New Annual Accomodation $= 31200 \times 1.035 = \$32292$

New Weekly Accomodation $= \dfrac{32292}{52} = \$621$

New Annual Transport $= 5980 \times 1.035 = \$6189.30$

New Weekly Transport $= \dfrac{6189.30}{52} = \$119.03$

Food and Groceries remain unchanged $= \$11\,440.00$

Annual Saving is decided to be $= \$5200$

Weekly saving $= 5200 / 52 = \$100.00$

Find new Annual Daily and Discretionary spending as amount remaining when all other spending is done

Amount $= 60229 - (\$32292 + 6189.30 + 11440 + 5200$

$= 60229 - 55121.30$

$= \$5107.70$

Weekly Daily & Disc $= \$\dfrac{4515.68}{52} = \$86.84$

Budget for This Year		
Sector	**Annual Amount**	**Weekly Amount**
Accommodation	\$32 292.00	\$621.00
Transport	\$6 189.30	\$119.02
Food and Groceries	\$11 440.00	\$220.00
Daily and Discretionary Spending	\$5 107.70	\$98.23
Saving	\$5 200.00	\$100.00
Total Income	\$60 229.00	\$1 158.25

4

4 marks for complete table, with any reasonable decimal rounding

3 marks for table with a minor error or omission

2 marks for three correct and relevant calculations or equal merit

1 mark for correct income or equal merit